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MILITARY SPECIFICATION

PANS, FRYING AND BRAISING, ELECTRIC TILTING TYPE

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers commercial electric, covered tilting-type frying and braising pans (hereinafter called frying pans) either floor-mounted or wall-mounted, and equipped with adjustable temperature controls.

1.2 Classification. Frying pans covered by this specification shall be of the following types, classes, styles, sizes, and models as specified (see 6.2.1):

Type I - With extension frames and trays.
Type II - Without extension frames and trays.

Class B - Noninsulated pan sidewalls.

Style A - Floor mounted.
Style B - Wall mounted with chair carrier assembly.
Style C - Wall mounted without chair carrier assembly.

Size 1 - 23 inches long by 23 inches wide, 7 inches pan depth.
Size 2 - 40 inches long by 23 inches wide, 7 inches pan depth.
Size 3 - 23 inches long by 23 inches wide, 9 inches pan depth.
Size 4 - 40 inches long by 23 inches wide, 9 inches pan depth.
Size 5 - 31 inches long by 23 inches wide, 7 inches pan depth.
Size 6 - 31 inches long by 23 inches wide, 9 inches pan depth.

Model A - Without casters.
Model B - With casters.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7310

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-C-77 - Casters, Rigid and Swivel (Institutional Duty).
QQ-A-225/6 - Aluminum Alloy Bar, Rod, and Wire; Rolled, Drawn or Cold Finished.

MILITARY

MIL-R-11337 - Ranges, Packaging and Packing of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A109 - Cold-Rolled Carbon Steel Strip.
A167 - Chromium-Nickel Steel Stainless and Heat Resisting Plate; Sheet and Strip.
A176 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet and Strip.
A240 - Stainless and Heat-Resisting Chromium and Chromium-Nickel Steel Plate, Sheet and Strip for Fusion-Welded Unfired Pressure Vessels.
A269 - Stainless and Welded Austenitic Stainless Steel Tubing for General Service.
A276 - Stainless and Heat-Resisting Steel Bars and Shapes.
A366 - Cold-Rolled Carbon Steel Sheet, Commercial Quality.

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A480 - Stainless Steel Plate, Sheet, Strip, General Requirements for.
A569 - Hot-Rolled Carbon Steel Sheets, Commercial Grade.
B221 - Aluminum Alloy Extruded Bars, Rods, Shapes and Tubes.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL SANITATION FOUNDATION (NSF)

No. 4 - Commercial Cooking and Hot Food Storage Equipment.
Listing of Food Service Equipment.

(Application for copies should be addressed to the National Sanitation Foundation, NSF Building, 3475 Plymouth, Ann Arbor, MI 48106.)

UNDERWRITERS' LABORATORIES, INC. (UL)

UL 197 - Commercial Electric Cooking Appliances.
UL 486A - Wire Connectors and Soldering Lugs for Use with Copper Conductors.
UL 486B - Wire Connectors for use With Aluminum Conductors.
Electric Appliance and Utilization Equipment List.
Electrical Construction Materials List.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The frying pan shall consist essentially of a pan section either hinged to tilt through a mechanical linkage (see 3.8.3.1) or trunnion-mounted to tilt by means of a hand wheel operating a worm-gear (see 3.8.3.2), a supporting framework, a hinged cover, and a control system. The mechanism shall be capable of rotating the pan 90 degree (°) from the normal horizontal operating position to permit draining and shall lock the pan in any position at which the crank or hand wheel is released. When a worm-gear is furnished, the pan shall be rotated at least 90° from the normal horizontal operating position.

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3.2 First article. When specified (see 6.2.1), the contractor shall furnish a frying pan of the classification specified for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Standard commercial product. The frying pan shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the frying pan being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Standards compliance. The frying pans shall meet the requirements of the following sanitary and safety standards:

a. NSF No. 4.
b. UL 197

3.4.1 Certification. Prior to approval of the first article, if one is submitted, or prior to approval of the first shipment, the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative that the frying pans he proposes to supply under this specification meet the requirements of the standard specified in 3.4. Acceptable evidence of meeting these standards shall be as follows:

a. UL 197 - A UL approved listing and listing mark.
b. NSF No. 4 - Listing of the model being furnished in the current edition of the NSF Listing of Food Service Equipment and authorized display of the NSF seal on the finished frying pans, or a letter of certification for the frying pans issued by NSF under its special one-time evaluation certification service for equipment being furnished under a Government contract.

3.4.2 Alternate certification. In lieu of the certification required under 3.4.1(a) and 3.4.1(b), acceptable evidence of meeting the requirements of the standards cited will be a certified test report from a recognized independent testing laboratory, selected by the manufacturer and approved by the contracting officer, or in the case of NSF certification, approved by the cognizant medical authority acting in an advisory capacity to the contracting officer, indicating that a frying pan identical to the items to be furnished under this specification has been inspected in accordance with the applicable standard and meets the requirements thereof. Alternate certification shall be submitted prior to approval of the first article or prior to commencing production unless frying pans will be furnished from the contractor's current inventory, in which case alternate certification shall be sumitted prior to the first shipment.

3.4.3 Applicability. Details for frying pan construction, design, and performance covered by this specification shall be supplementary to the requirements of the standards cited in 3.1 and compliance with and approval under these standards shall not relieve the contractor of the responsibility

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for meeting the requirements of this specification. Definitions of terms given in NSF No. 4 shall be applied as required in interpreting and implementing the requirements of this specification.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.6.1 Steel.

3.6.1.1 Stainless. Stainless steel shall conform to the requirements of ASTM A167, A176, A240, A269, and A276. Minimum thickness and tolerance of sheet steel shall conform to the requirements of ASTM A480 for the gage specified. Nominal thicknesses as stated herein are included for reference.

3.6.1.2 Carbon steel. Carbon steel used in the fabrication of the frying pans shall conform to the applicable requirements of ASTM A109, A366, or A569.

3.6.2 Aluminum. Aluminum used in the fabrication of the frying pan shall conform to the applicable requirements of ASTM B221.

3.6.3 Hardware and fittings. All cast or formed hardware and fittings shall be stainless steel, nickel or chromium-plated steel, brass, or bronze in accordance with the manufacturer's standard practice.

3.6.4 Fasteners. Bolts, screws, nuts, or other fasteners which must be removed to disassemble the equipment for installation or repairs, or to secure stainless metal parts, shall be stainless steel, nickel, or chromium-plated steel.

3.7 Design. The equipment shall be so designed and constructed so as to allow installation of a continuous row of adjacent units and to permit ordinary repairs or replacement of control units and thermostats to be made. Means to prevent the pan section from rotating backwards in the event the control handle is not properly used shall be provided. The equipment shall be so designed that during normal operation the manual parts move readily and the temperature of handles, actuating members of switches, and other frequently handled surfaces do not exceed 120° Fahrenheit (F) when tested as specified in 4.5.1.2. Temperature distribution over the frying pan cooking surfaces shall be uniform to the extent that, with simultaneous temperature measurements at

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various points on the cooking surface (see 3.9.1 and figure 1), each of the temperatures measured at the peak of the temperature cycle shall be within ±14 percent of the average temperature measurements of all the points. The frying pan shall have a minimum rating in kilowatts of 7.5 for size 1 and 3 pans, 12 for size 5 and 6 pans, and 15 for size 2 and 4 pans. Full power shall be delivered at the voltage specified (see 3.10.3).

3.8 Construction.

3.8.1 Dimensions. The pan depth specified in 1.2 shall be the perpendicular distance between the pan cooking surface and the lip of the pan with a tolerance of ±1/2 inch. Cooking surface dimensions specified in 1.2 shall have a tolerance of ±2 inches. The height of the unit measured from the lip of the level pan to floor shall be 35 to 41 inches for 9-inch pans and 34 to 41 inches for 7-inch pans. Each leg of style A pans shall be provided with a means of vertical adjustment not less than 1 inch to level the unit.

3.8.2 Framework and mounting. Unless otherwise specified (see 6.2.1), the framework of style A frying pans shall be constructed of pipe or tubing not less than 1-1/2 inches in outside diameter and at least 0.060-inch thick, and shall be of stainless steel. Framework and members shall be of stainless steel conforming to ASTM A167, A269, or A276 and shall meet the requirements of NSF No. 4. When specified (see 6.2.1) frame members of aluminum alloy shapes shall meet the requirements of ASTM B221. Structural members and surfaces shall be treated and finished in accordance with NSF No. 4. The frame of style A, B, or C frying pans shall be designed and constructed to support the pan when tested in accordance with 4.5.7. Style A, model A frying pans shall be furnished with stainless, adjustable steel feet. The frame of style A, model A pans shall be designed and constructed to adequately support the pan when tested as specified in 4.5.4. Style A, model B frying pans shall be furnished with 5-inch diameter casters conforming to class B or C, style C of FF-C-77. Casters and caster-mounting areas shall show no evidence of damage, deterioration, or failure when tested as specified in 4.5.5. Unless four swiveling casters are specified (see 6.2.1), two of the casters shall be rigid and two shall be swiveling. The swiveling casters shall be on the operator's side of the pan with foot-operated brakes or locking devices which shall show no movement when tested as specified in 4.5.6. The style B frying pan shall have the same framework as style A, except that it shall provide for wall mounting with a chair carrier assembly. The style C frying pan shall have the same frame as style A, except that it shall provide for wall mounting without a chair carrier assembly.

3.8.2.1 Enclosures. Enclosures shall meet the sanitation, material, and fabrication requirements of NSF No. 4. Cabinet-type bearing and control enclosures shall be seamless construction and shall be furnished with removable covers or panels to permit access to internal components. When cabinet-type enclosures are specified for styles B or C, the enclosures shall be strong enough to support the full weight of a loaded pan and thereby serve as the wall-mount attachment. On trunnion-type fry pans, electrical controls and signal lights shall be housed in a stainless steel sheet cabinet-type enclosure on one side of the unit. At the option of the manufacturer, the tilting mechanism shall be housed either in a standard aluminum alloy or dairy bronze gear box or in a stainless steel sheet cabinet-type enclosure. Enclosures fabricated of stainless steel shall be not less than 0.050-inch thick.

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3.8.3 Tilting system. The frying pan tilting system shall be as specified in 3.8.3.1 or 3.8.3.2 (se 6.2.1). The tilting system shall provide a means of manually or electrically rotating the loaded or empty pan from the normal horizontal operating position to the fully tilted position. The frying pan shall be designed to provide positive natural drainage of fluids to the pouring lip. The tilting mechanism shall be self-locking with the loaded or empty pan in any position between level and fully raised when the operating crank or wheel or switch is released, and shall be provided with a means to prevent backward rotation of the pan past the horizontal position.

3.8.3.1 Mechanical linkage. Fry pans furnished with a tilting mechanism having a mechanical linkage shall be constructed with the rotation mechanism enclosed to conform to NSF No. 4. Shafts or linkages which extend through bearing or sanitary enclosures shall be provided with suitable seal at the opening to prevent intrusion of washing solutions onto lubricated areas or the migration of lubricant, dirt, and contaminants from the mechanism area onto outer surfaces. If the operating crank or wheel is removable, the socket assembly shall meet this sealing requirement with the crank installed or removed. The operating crank or handwheel shall be stainless steel, nickel-plated steel, or nonferrous metal. Nonmetallic accessories forming a part of cranks or other operating controls shall be capable of withstanding normal abuse when the frying pan is at operating temperatures, and shall be of a material conforming to NSF No. 4.

3.8.3.2 Worm-gear and trunnion linkage. The frying pan tilting system shall consist of a worm-gear arrangement wherein rotation of the pan is affected through a handwheel-operated solid drive shaft to the trunnion shaft. The gear system shall be capable of rotating the pan is affected through a handwheel-operated solid drive shaft to the trunnion shaft. The gear system shall be capable of rotating the pan 90° from the horizontal position and shall be self-locking at any position within this arc. The solid drive shaft shall be stainless steel having a minimum diameter of 0.750 inch or a minimum 1-1/4 inch diameter aluminum alloy shaft that conforms to QQ-A-225/6, temper T6 or T851. The shaft shall extend from the front of the bearing enclosure with a suitable seal provided at the enclosure opening. The drive shaft shall be fitted with a chrome-plated or stainless steel handle, permanently attached to the drive shaft.

3.8.4 Frying units. Frying units shall be welded, seamless drawn, or cast. Projecting or shaped pouring spouts shall be formed, drawn, or welded. A heat sink shall be provided under the heated cooking surface. Attached heat sinks shall be clamped or bolted in a manner to maintain uniform efficient heat transfer over the entire surface. Integral-clad heated surfaces shall be constructed of material not less than 15 percent clad and shall be not less than 0.375-inch thick with an interior stainless steel surface not less than 14 gage (0.078-inch nominal) with or without a copper-clad inter-layer at the heat sink. Stainless steel cooking surfaces shall have a smoothness of not less than 100 grit finish. All cooking surfaces shall be free of porosity, foreign matter, cracks, and dishouts. All internal corners and angles on food contact surfaces shall conform to NSF No. 4. A suitable pouring lip shall be formed in the front wall of the pan, or a lip or spout shall be separately formed and welded to the front wall to facilitate complete drainage when the pan is tilted. The separately-formed spout shall be constructed of the same material

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and the same or heavier gage as the pan. The spout shall not leak when the pan is tilted for drainage.

3.8.5 Heating elements. Heating elements shall be of the enclosed coil, strip or manufacturers designed type attached to the heat sink of the pan, uniformly distributed in a manner to give good thermal contact, yet allow movement for expansion and contraction. The bottom section of both class A and B pans shall be enclosed by a heat insulating shield consisting of 1-inch thick glass fiber or equivalent insulation. Able to withstand surface temperatures up to 450°F without damage. A removable stainless steel pan not less than 0.050-inch thick shall enclose the element heat shield. Terminals of heating elements shall be placed as far from the grease and heat zone as practicable. Each terminal shall be effectively sealed to prevent moisture or grease from entering the terminals or heating elements.

3.8.6 Pan enclosure. Pan enclosure shall be constructed as per manufacturers standard design.

3.8.7 Cover. A hinged cover shall be furnished on each unit. The cover shall be designed to completely cover the pan including the pouring spout and shall be die-drawn or fabricated without seams of stainless steel not less than 0.050-inch thick. The cover shall be hinged to move independently of the motion of the pan, and shall be hinged to the stationary framework or to the right and left stationary enclosure console boxes. All welds shall be ground smooth and polished. The cover shall be spring or otherwise counter-balanced for ease of operation, and shall be designed by means of stops or other devices to remain in a partially or fully opened position. Stops to prevent the cover from falling backward shall be provided. Any device depending on friction to hold the cover open shall not be acceptable unless it is fully protected from possible exposure to grease and oil and is readily adjustable manually. Covers shall fully comply with NSF requirements, shall be of durable design, and shall not flex or distort during normal operations. The cover shall be provided with an adjustable vent to prevent condensation from forming over the food product. Covers, when open shall be designed to prevent from draining on the floor.

3.8.7.1 Lift bar. Covers shall be provided with a stainless steel lift bar designed to meet all requirements of 3.7 and permit operating the cover in the fully open, intermediate, and closed positions to minimize reaching across hot cooking vapors.

3.9 Performance.

3.9.1 Temperature distribution. At the peak of the cooking cycle as indicated cut-off of the primary thermostat, temperatures distributed over the dry frying pan cooking surface in accordance with figure 1 (see 4.5.1.1) shall be uniform within ±14 percent of the average measured temperature of all the points (see 4.5.1.2).

3.9.2 Thermal efficiency. The pan containing at least 4 inches of water covering the cooking surface and provided with the rated power shall have a thermal efficiency of not less than 70 percent when tested as specified in 4.5.3, at the highest temperature setting.

3.9.3 Energy input. With rated power provided, the empty pan shall be capable of reaching operating temperature in not more than 15 minutes (see 4.5.7).

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3.10 Controls. Controls shall be located in the enclosure specified in 3.8.2.1. The unit shall be regulated by means of an ON-OFF switch connected to a signal light and by two thermostats. It shall be acceptable to combine the switch and primary thermostat in one control. The switch and thermostats shall control a magnetic contactor suitable for the rating and power of the pan. The magnetic contactor assembly furnished shall be entirely within the specified enclosure. The controls shall be so located as to be readily accessible for operation from the front of the frying pan. The signal light shall indicate when the heating elements are energized. The primary thermostat shall be of the variable type, and shall be so calibrated and so located as to be capable of maintaining in the test cooking surface (see 4.5.2), the desired temperatures ranging from an average low of 200°F to an average high of 400°F. The thermostat shall be designed to regulate the cooking surface temperatures throughout the entire range of thermostat settings. Unless otherwise specified (6.2.1), the secondary thermostat shall be of the manual or automatic resetting type and shall have its sensing element located on the pan body.

3.10.1 Signal lights. The frying pan shall be furnished with signal lights to indicate that heater elements are energized. At the option of the manufacturer, signal lights shall be provided to indicate that power is ON or other operation conditions as applicable.

3.10.2 Thermostats.

3.10.2.1 Primary thermostat. An adjustable thermostat shall be furnished capable of controlling temperature distribution on the cooking surface to a maximum of 425°F. The thermostat shall be adjustable over a range of 200°F to 400°F. The thermostat shall have a positive "OFF" position which disables the circuit and be marked with graduations at 25°F intervals.

3.10.2.2. Secondary thermostat. The secondary thermostat shall be wired in series with the primary thermostat, to disable the power circuit when temperture at the sensor located in the pan body exceeds 460°F. Reset of the power switch shall re-energize the circuit and restore the pan to operation once the temperature at the sensor reaches a level below 460°F.

3.10.3 Electrical components. All wiring used on the frying pan shall be enclosed and well protected and conform to NFPA No. 70. Wiring insulated with asbestos or other fibrous material shall be protected against the intrusion or absorption of grease. All wires to the frying pan shall remain free of tension throughout the full arc as the frying pan is rotated between the fully-lowered to the fully-raised position. Wiring between terminal points shall be enclosed or clamped securely with stainless or electroplated metal fasteners. Unless otherwise specified (6.2.1), the frying pans shall be wired for 208 volt (V), 3 phase, 4 wire, 60 Hertz power and the thermostat shall be wired with not more than 208V throughout the control circuit.

3.10.4 Electrical connectors. Electrical connections shall be screw type or bolt type meeting UL 486A and 486B as applicable. Lead wires shall be crimped to connectors using UL recognized mechanical crimping tools to assure permanent low-resistance connections. Connectors shall be treated, plated, or stainless metal, and shall be flattened at terminal contact areas to assure uniform contact. Leads to terminals not easily identifiable shall be stamped

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or identified with a suitable coding and terminals shall be correspondingly stamped or marked for easy replacement or repair.

3.11 Finish. All exposed stainless sheet steel surfaces shall have a commercial No. 2B or better finish with no stains, foreign matter, bulges, or dents. The surface finish of the cooking surface shall be as specified in 3.8.4. Cold-rolled steel finish shall be as rolled and free of visible scale.

3.12 Identification marking. Each frying pan shall be legibly and permanently marked with the following information:

a. Manufacturer's name or trademark.
b. Manufacturer's model number or equivalent.
c. The power rating and other data as specified in UL 197.

3.13 Extension frames and trays. Type I frying pans shall be furnished with trays or pans fabricated of stainless steel not less than 18 gage (0.050-inch) conforming to NSF No. 4. Two side-mounted extension frames and trays or one front-mounted self-leveling extension frame and tray shall be furnished with the frying pan as required in 1.2.

3.14 Commercial publications. The manufacturer's standard commercial publications (operator, maintenance, and parts manuals, as available) shall be furnished with each frying pan as specified (see 6.2.1 and 6.3).

3.15 Workmanship.

3.15.1 Stainless steel fabrication. The stainless steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the stainless steel to a value less than intended by the design. All bends shall be made by controlled means to insure uniformity of size and shape.

3.15.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.15.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.15.4 Aluminum fabrication. Aluminum shall have a uniformly clean appearance free from all fabrication and mill markings, smudges, and foreign material. All exposed surfaces and edges shall be smooth and free from sharp projections.

3.15.5 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts

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connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.15.6 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that supplies and materials are inspected for compliance with all the requirements specified herein and in applicable referenced documents.

4.1.2 Standards compliance. The contractor shall make available to the contracting officer or his authorized representative evidence of compliance with the applicable standard(s) cited in 3.4.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one frying pan when a first article is required (see 3.2 and 6.2.1 and 6.4). This inspection shall include the examination of 4.4 and the tests of 4.5.1 through 4.5.7. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the tests of 4.5.4 through 4.5.7, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. All frying pans of the same classification offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection

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5.2.1 Level A. Ovens preserved and packaged as specified in 5.1.1 shall

shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.3.1 Sampling for examination. Examination shall be based on inspection level II and an Acceptable Quality Level (AQL) of 1.5 defects per one hundred units.

4.3.2 Sampling for tests. Sampling for the operational test of 4.5.7 shall be 100 percent. All other tests shall be based on inspection level S-4 and an AQL of 1.5 defects per one hundred units.

4.4 Examination. Each frying pan shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements shall constitute a defect.

4.5 Tests.

4.5.1 Performance tests.

4.5.1.1 Preparation for performance test. For the test of 4.5.1.2 and 4.5.2, thermocouples having a temperature range suitable for the purpose shall be placed in the sample frying pan on the horizontal cooking surface. The number of thermocouples used shall depend on the size of the cooking surface, as shown in figure 1. Five of the thermocouples shall have fixed locations on the horizontal cooking surface and additional thermocouples, when used, shall be placed at random locations. In lieu of thermocouples, a surface pyrometer may be used to record cooking temperatures. The temperatures shall be read and recorded in sequence, as quickly as possible. Pans designed to provide a cool edge to permit drainage of fats and gravies without scorching shall have this cool edge excluded from the area shown in figure 1.

4.5.1.2 Conduct of performance test. The frying pan shall be heat-soaked at ambient room temperature of 75° ±5°F for at least 2 hours. With the thermostat set at 375°F, the empty pan shall be operated with the cover closed at the rated power supply until the unit has stabilized at the desired operating temperature as evidenced by cycling of the unit at least five times under control of thermostat. The unit will be permitted to cycle an additional five times, and the temperature measured to determine compliance with 3.9.1. During the test, the temperature of handles, actuating members of switches, and exposed surfaces which the operator may contact shall be checked to determine compliance with 3.7. Any nonconformance to 3.7 or 3.9.1 shall constitute failure of this test.

4.5.2 Thermostat calibration and control amplitude. With the pan in the horizontal position, the empty pan shall be operated at 200°F, at 325°F, and at the maximum setting through a minimum of 10 cycles each and the temperatures measured (see 4.5.1.1) to determine compliance with 3.10 and 3.10.1. Average surface temperature, measured by surface thermocouples or by surface pyrometer at the locations indicated in figure 1, shall be within ±8°F of the dial setting. The temperature amplitude, from heaters-on to heaters-off, shall not exceed 40°F. The cover of the pan shall remain closed

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when practicable during the test. Any nonconformance to 3.10 or 3.10.1 shall constitute failure of this test.

4.5.3 Thermal efficiency test. Sufficient water having a temperature of 60° ±5°F shall be added to fill the pan to a depth of 4 inches ±1/8 inch, then the water shall be removed, weighed, and stored for future use. Additional water shall be added and the frying pan shall be operated at the highest setting, with the rated power supply provided, until the water boils for a sufficient time for the pan and contents to reach an equilibrium temperature condition. The pan shall then be emptied and the stored water added. Refilling shall be accomplished as quickly as practicable to minimize heat changes to the pan or surrounding area. The pan shall be operated as specified above. When the water reaches a temperature of not less than 200°F, the power supply shall be turned off and the electrical energy in kilowatt hours (kWh) consumed shall be determined. The thermal efficiency of the pan shall be determined using the formula:

$$E = \frac{W \times (Q_2 - Q_1)}{C \times kWh}$$

Where W = weight of water, lb.
Q_1 = initial temperature of water, F.
Q_2 = final temperature of water, F.
C = factor for $\frac{Btu}{kWh} = 3412$
kWh = power consumed.

The cover of the pan shall remain closed as much as practicable during the test. Temperature shall be determined using a thermometer immersed with the sensing bulb at 2 inches ±1/4 inch above the cooking surface. Any nonconformance to 3.9.2 shall constitute failure of this test.

4.5.4 Stand test. The style A, model A pan shall be dragged by the lower end of each leg in turn across a level concrete floor for not less than 3 feet to determine compliance with 3.8.2. Examination shall reveal that the pan is free from damage or permanent deformation with special attention to dislodgement of trunnions on trunnion-mounted designs. Any nonconformance to 3.8.2 shall constitute failure of this test.

4.5.5 Obstacle test. The style A, model B pan with casters, loaded to capacity, shall be moved across an obstacle not less than 1/8-inch high nor more than 3/8 inch high at least 10 times at a speed not less than 2 miles per hour to determine compliance with 3.8.2. Any nonconformance to 3.8.2 shall constitute failure of this test.

4.5.6 Brake test. The style A, model B pan with casters, loaded to capacity, shall be placed on a 10 percent slope for not less than 10 minutes with the brakes set to determine compliance with 3.8.2. Any nonconformance to 3.8.2 shall constitute failure of this test.

4.5.7 Operational test. Each assembled pan shall be subjected to the following operational tests:

A. The pan shall be operated at the rated power supply with the thermostat at the highest setting. While the pan is heating, an examination shall be made to determine that all heaters are

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functioning and that uniform heating exists across the entire cooking surface. After a period of time not greater than 15 minutes, the surface temperature shall be measured and a check made to determine that the thermostat has cycled at least once and that the frying pan has reached operating temperature.

B. The pan shall be loaded with water to within 1 inch of the top or lip. All switches, controls, and moving parts shall be checked for fit, function, and smoothness of operation when the fry pan is cool and at the highest operating temperature to determine compliance with 3.7.

Any nonconformance to specified requirements shall constitute failure of this test.

4.6 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-R-11337.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-R-11337 with the level of preservation and the level of packing as specified (see 6.2.1).

6. NOTES

6.1 Intended use. Frying pans covered by this specification are intended for use at Military messes and other mass-feeding establishments to braise, simmer, boil, stew, and pan fry (except the type of deep-fat frying normally done by french-frying cookers).

6.1.1 Style. Where style B is specified, provisions should be made for instructions from the manufacturer regarding installation of the chair carrier to include reinforcement or modification of the wall structure, if required. Style C should be specified for shipboard use or for use in installations having steel reinforced walls specifically applicable for concentrated loads.

6.2 Ordering data. Acquisition documents should specify the following:

6.2.1 Acquisition requirements.

a. Title, number, and date of this specification.
b. Type, class, style, size, and model required (see 1.2).
c. When a first article sample is required for inspection and approval (see 3.2, 4.2.1, and 6.4).
d. When the type of framework construction is other than as specified, when frame members are constructed of aluminum alloy, and when four swiveling casters are required (see 3.8.2).
e. When a mechanical linkage or a worm-gear and trunnion linkage tilting mechanism is required (see 3.8.3).
f. When a manual reset secondary thermostat is required (see 3.10).
g. When electrical power other than as specified is required (see 3.10.3).
h. When commercial publications and detailed requirements are required (see 3.14).
i. Level of preservation and level of packing required (see 5.1).

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6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved DD Form 1664, Data Item Description (DID), and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph No.	Data requirements title	Applicable DID No.	Option
3.14	Publications, Commercial	DI-M-24010A	

(DIDs related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, Acquisition Management Systems and Data Requirements Control List. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item consisting of one complete frying pan, or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one complete frying pan. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - GL
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Air Force - 99

Preparing Activity:

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(Project 7310-0679)

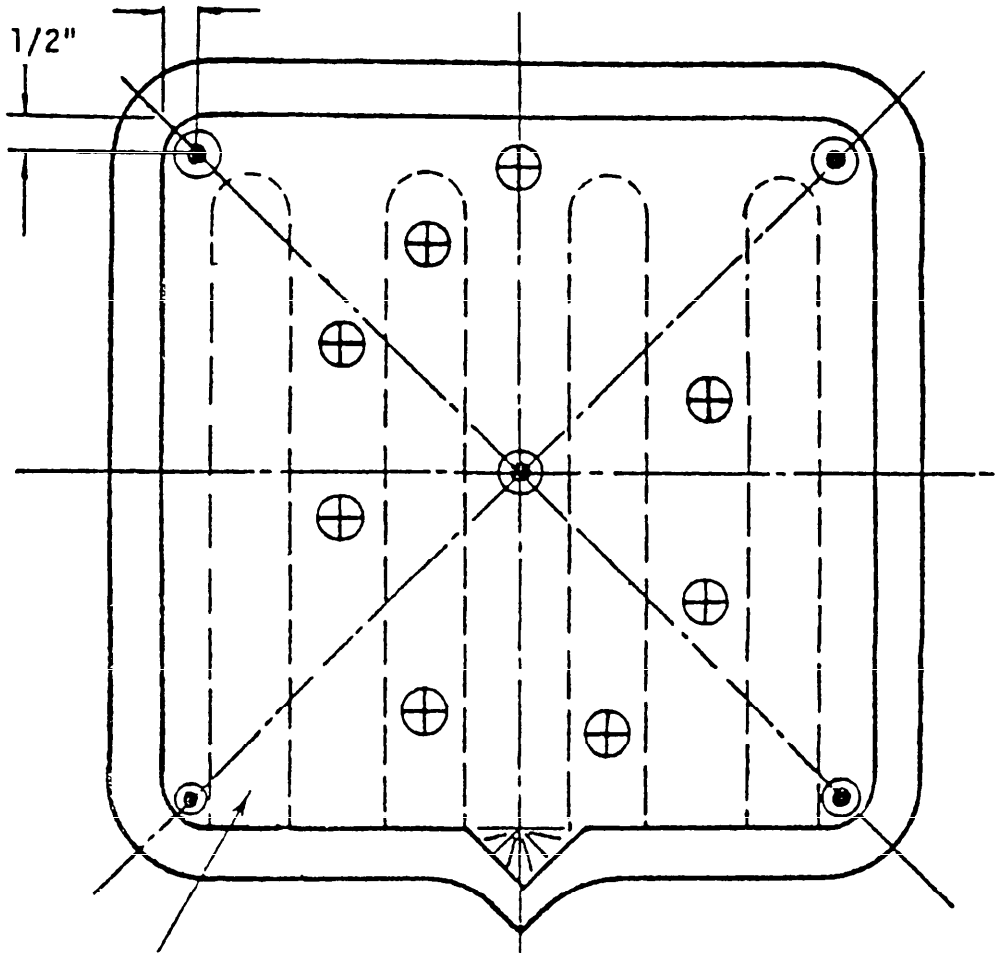
Review Activities:

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Navy - MS, MC, SA
DLA - GS
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User Activity:

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⊙ Fixed thermocouple locations shall include one center location and the four corner locations as close as practicable, but in no case less than 1/2 inch to the edge of the flat cooking surface.

⊕ Random thermocouple locations in section bounded by diagonals shall not exceed the number of thermocouple locations in any other section by more than one and shall be in accordance with 4.5.1.1.

FIGURE 1. Thermocouple locations.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-23694E	PANS, FRYING AND BRAISING, ELECTRIC TILTING TYPE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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